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Spectrum and Energy Efficiency in CRNs Using Dynamic Channel Reservation and Komodo Mlipir Optimization

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ABSTRACT

Cognitive Radio Networks (CRNs) block new arriving services and remove existing services to increase the spectrum efficiency. Particularly, Secondary Users (SUs) experience a decline in service deprivation in the arrival of Primary Users (PUs). To overcome this, a Spectrum and energy Efficiency in Cognitive Radio Networks using Dynamic Channel Reservation and Komodo Mlipir Optimization (SE-KMOA-CRN) is proposed. The proposed method splits reserved and non-reserved bands from the accessible spectrum. Numerous channels from the non-reserved band are dynamically assigned to the reserved band to support services that experience disruptions when functioning on non-reserved band. From the non-reserved band, a count of channels dynamically assigned to the reserved band to support services that experience disruptions when functioning on the non-reserved band. It enables priority-based channel assignment and termination while providing dynamic access to the obtainable spectrum. Here, Allocation of channels to Reserved Band CRN using Dynamic Channel Reservation Algorithm (DCRA). Komodo Mlipir Optimization Algorithm (KMOA) is employed for Service selection and Channel restoration. The proposed SE-KMOA-CRN method is modelled utilizing Continuous Time Markov Chain (CTMC) with mathematical equations for some quality of service (QoS) factors is obtained. The SE-KMOA-CRN approach attains higher Throughput, lower End-to-End Delay, higher Energy Efficiency, greater spectral efficiency and higher Packet Delivery Ratio compared with existing methods, like Multiple objective optimization for spectrum along energy efficacy trade-off in IRS-assisted CRNs using NOMA (SE-MOO-CRN), Energy efficacy in CRN utilizing cooperative spectrum sensing under hybrid spectrum handoff (SE-TOA-CRN) and Energy-effectual cross-layer spectrum sharing on CR green IoT (SE-CRG-CRN) respectively.

KEYWORDS

Cognitive radio network); Dynamic channel reservation; Energy efficiency (EE; Komodo Mlipir optimization

1. INTRODUCTION

The idea of cognitive radio helps to develop a new generation of networks by providing radio equipment with intelligence learning and decision-making skills. This allows radio equipment to detect the state of its surroundings and adjust its operating settings accordingly [1–3]. The ability of intelligent radio to dynamically access spectrum radio is the most widely used in the literature [4]. Due to the Federal Communications Commission's (FCC) spectrum resource constraint and growing user base, unlicensed terminal users can now utilize these resources by modifying their settings to match the new frequency bands [5]. Cognitive radio technology allows unlicensed users (secondary users, SUs) to transmit in licensed frequency bands to efficiently utilize the spectral band without affecting the quality of service (QoS) of licensed users' (primary users, PUs) transmission [6].

CRNs are designed to improve spectrum utilization by allowing Secondary Users (SUs) to access underutilized bands assigned to Primary Users (PUs), but this often results in service disruptions for SUs when PUs reclaim their spectrum, leading to reduced Quality of Service (QoS) [7]. Existing methods for managing spectrum and energy efficiency are insufficiently dynamic and adaptable, causing frequent inefficiencies and interruptions. This research is motivated by the need to develop a more robust solution that can dynamically allocate and restore channels, ensuring continuous service for SUs and optimizing spectrum efficiency. The proposed SE-KMOA-CRN aims to address these gaps by introducing advanced, adaptive algorithms for better performance and reliability in CRNs. This research proposes a unique approach, namely effective Energy and Spectrum Utilization (ESU), to improve network throughput by making effective use of the available spectrum. The two

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algorithms that make the scheme are Enhanced KMOA and DCR. While KMOA permits service selection and channel restoration, DCR balances channel assignment among the reserved and non-reserved bands. ESU is considered a hybrid approach including elements of both channel and no channel reservation. The scheme is represented as CTMC. The main contributions of this research are abridged below.

- The proposed technique called ESU, which combines DCRA and KMOA algorithms, is efficient with spectrum.
- DCRA determines dynamically the amount of reserved channels for all service arrival or leaving. Numerical outcomes depict that DCRA has a constant time complexity and more time-efficient than the baseline scheme when it comes to spectrum reservation.
- The performance evaluation uses the combined consequences of channel failure, channel restoration, channel non-accessibility, and the possibility of forced termination.
- Channel access, capability, retention capacity, and possible network unavailability for services are all expressed in a closed form.

The continuing paper is organized as follows: the literature survey is presented in Portion 2, the proposed approach is labeled in Portion 3, outcomes are proved in Portion 4, and conclusion is depicted in Portion 5.

2. LITERATURE SURVEY

Several research studies were previously suggested in the literature depending on spectrum and Energy Efficient in CRN; a few works are reviewed here. Table 1 depicts the comparison of the literature survey.

Zheng, *et al.*, [8] presented Spectrum utilization development for multiple channels EH-CRN with spectrum sensing. Here, in the multiple channel EH-CRN, the optimization of the longer-term SUE is formulated as a Markov decision process (MDP) problem through the infinite size of system space and action space. An SS, channel selection, EH, and data transmission (SCED) scheme is suggested that adaptively determines the SS and EH operations of the K-antenna ST depending on the residual energy and sensing outcomes. It attains high throughput but the energy efficiency was reduced.

Pankajavalli, *et al.*, [9] presented Qos-Aware Resource Allocation and Spectrum-Efficient User Allocation with Enhanced Uplink Transmission in U-LTE Networks Co-occurrence with Wi-Fi by CRN. The suggested paper focuses on the way the Flower Pollination optimization Algorithm (FPA) improves the coexistence of Wi-Fi along LTE networks utilizing Cognitive Radio (CR). This framework's primary goal is to make SUs with excellent quality of service (QoS) possible at reasonable costs and data rates in accordance with all networks that are accessible but have inactive systems' interruption constraints.

Balachander, *et al.*, [10] suggested a cooperative spectrum sensing for 5G wireless communication in cognitive radio networks for the Internet of Things. In the presented paper, Offset Quadrature Amplitude Modulation Universal Filtered Multicarrier Non-Orthogonal Multiple Access techniques were used to provide an optimal CSS for CRN. In IoT-based 5G communication, the ideal CRN solution offers the best possible bandwidth. PAPR was removed with the Energy Efficient All-Pass Filter. The deployment strategy enhances system throughput, energy efficiency, and reliability while maintaining Quality of Service (QoS).

Table 1: Comparison of Literature Survey

Name of Author	Objective	Performance Metrics	Merits	Demerits
Zheng, *et al.*, [8]	Spectrum utilization improvement for multi-channel EH-CRN with spectrum sensing	Throughput, Delay, Energy Consumption, and energy efficiency.	Lesser delay.	Lesser throughput.
Pankajavalli, *et al.*, [9]	Spectrum-Efficient User along Qos-Aware Resource Allocation with Enhanced Uplink Transmission in U-LTE Networks Co-occurrence with Wi-Fi by CRN	Energy efficiency, sensing performance, handoff time, miss-detection, and throughput,	Greater throughput.	Lesser energy efficiency.
Balachander, *et al.*, [10]	Cooperative Spectrum Sensing Deployment for the Cognitive Radio Networks for the Internet of Things 5G Wireless Communication	Throughput, Delay, Energy Consumption, and energy efficiency.	Lesser delay.	Greater Energy Consumption.
Wu, *et al.*, [11]	Multi-objective optimization for spectrum and energy efficiency tradeoff in IRS-assisted CRNs with NOMA	Throughput, spectrum efficiency, energy efficiency, and packet delivery ratio.	Higher packet delivery ratio.	Lesser spectral efficiency.

Wu, *et al.*, [11] presented multi-objective optimization for the trade-off among spectrum and energy efficiency in CRNs using IRS assistance and NOMA. The downlink multiple-input-single-output (MISO) CRNs with NOMA utilize an intelligent reflecting surface (IRS). A multiple objective optimization (MOO) structure was established under the perfect and imperfect channel state information (CSI) since the tradeoff between SE and EE was significant. The spectrum efficiency was increased and throughtput was reduced.

3. PROPOSED METHODOLOGY

In this section, SE-KMOA-CRN is proposed. The SE-KMOA-CRN method addresses the challenges faced by Secondary Users (SUs) in CRNs due to the arrival of Primary Users (PUs) by employing DCRA and KMOA. DCRA dynamically adjusts a count of reserved channels based on real-time traffic, prioritizing PU access while ensuring SUs maintain service continuity, thereby reducing interruptions and blocking probability. KMOA optimizes service selection and channel restoration, quickly reallocating pre-empted SUs to newly available channels, and minimizing forced terminations and downtime. This approach enhances overall spectrum efficiency, throughput, and energy efficiency, while reducing end-to-end delay, thus significantly improving the QoS for SUs in the network. Figure 1 portrays the block drawing of SE-KMOA-CRN. The comprehensive discussion of SE-KMOA-CRN is given below.

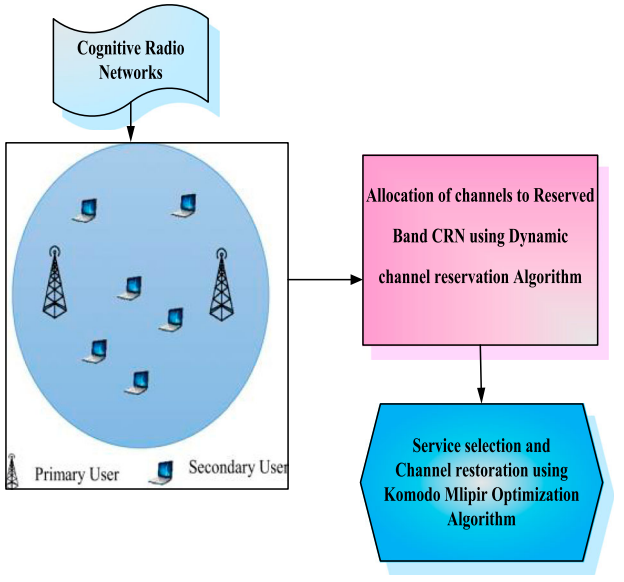


Figure 1: Block diagram of the proposed SE-KMOA-CRN system

3.1 System Model

An infrastructure-basis CRN design, including centralized basestation, is considered in this work. Every user who falls inside the coverage area of the BS is under its authority. A primary and a secondary network coexist. The Primary Network (PN) is made up of numerous PUs managed by BS and each uses its licensed spectrum. PUs along SUs are regulated through their respective base stations in CRN. The remaining part of the concept is comparable to the standard CRNs. SUs opportunistically search, locate, and reprocess optimum channels for the permitted spectrum allotted to PN. Channel scans might flop for a variety of reasons, which can interfere with ongoing conversations. Numerous are reserved in R-CRN, indicated by B, in R-CRN. The dimensions of B changes are based on the amount of channels used in the network. The restricted band is only accessible to interrupted PU and recently arriving SU that are unable to identify unoccupied channels in Non-Reserved CRN (NR-CRN). To prevent reserving an excessive count of reserved channels then increase the block chance for the entry of new services. A constraint indicated as $B_{\max}$ is placed on a count of reserved channels, here $B \leq B_{\max} \in A^{+}$. The process used for adapting B to modifications in the network environment is detailed in the following section.

3.1.1 Network Capacity

To determine the network capacity (NC) first assign the allocation of spectrum band (resource) including binary indicator is specified as $x_{w,l}^{m} \in \{0, 1\}$. The value $x_{w,l}^{m} = 1$ when the spectrum band is allocated as m^{th} allocation and the lth secondary user along with the wth Cognitive Base Station (CBS). The equation for e Signal to interference noise ratio (SINR) along with l^{th} the secondary user provided through the wth CBS occupies the mth spectrum band and the equation is given in Equation (1)

$$\zeta_{w,l}^{m} = \frac{g_{w,l}^{m}, T_{w,l}^{m}}{\sigma^{2} + N_{l}^{m}} \tag{1}$$

where $\zeta_{w,l}^{m}$ denotes the spectrum band, $g_{w,l}^{m}$ implies the channel gain among the wth CBS and l^{th} is the secondary user at the mth spectrum band, $T_{w,l}^{m}$ is the transmission power of wth CBS is allocated with the lth secondary user at m^{th} spectrum band, σ^{2} is the noise power, and the N_{l}^{m} is the interference encountered in the spectrum band. The network capacity attained with every secondary user in the wth CBS over the m resource with the objective function is formulated and its equation is given

in (2)

$$objectivefunction_1\,(NC) = \sum_{m\in M}\left(\frac{BW}{m}x_{w,l}^{m}\log_2(1+\zeta_{w,l}^{m})\right) \tag{2}$$

where BW is the bandwidth, Equation (2) is the objective function of the network capacity equation and to enhance the network capacity of the parameter ζ is optimized using the dynamic channel reservation scheme, therefore, efficiently assigning the channels in the spectrum.

3.2 Allocation of Channels to Reserved Band CRN Using the Dynamic Channel Reservation Algorithm

In this segment, the dynamic channel reservation algorithm reserving an optimal number of channels for allocating SU in the correct position and assigning the channels efficiently is discussed [12, 13]. Here ζ optimizes the parameter from Equation (2) for improving the network capacity; therefore, the spectrum and energy efficiency are improved. DCRA is a crucial component in optimizing the allocation of channels in CRNs. Its primary goal is to dynamically reserve an optimal number of channels for Secondary Users (SUs) while efficiently managing spectrum resources. The DCRA effectively manages dynamic spectrum allocation by adjusting the number of reserved channels based on real-time traffic conditions, thereby optimizing network capacity and spectrum efficiency. It handles the arrivals and departures of PUs and SUs efficiently, mitigates the impact of channel failures through spectrum handover, and reduces the likelihood of service interruptions and forced terminations. The stepwise processes for maximizing channel capacity are specified below:

***Step* 1:** Initialization

The initial factors of DCRA are initialized for improving network capacity. In this, the total number of channels in the CRN is represented as the N, and the blocking probability of the SU is represented as Pb_{bp}^{SU} is decreased by lessening the number of reserving channels RC at the time of traffic increases in the network. The traffic arrival rate of PU is represented β_{PU} and the traffic arrival rate of the secondary user is represented as β_{SU}. The objective function is used to improve the network capacity ζ and expressed in Equation (3)

$$\zeta_{NC} = \frac{N}{CA} \tag{3}$$

***Step* 2:** PU arrival

When a PU arrives in CRN, DCRA swiftly assesses the network's channel availability and dynamically adjusts the number of reserved channels to accommodate the PU's needs. By efficiently allocating idle channels to incoming PUs, DCRA minimizes the probability of service interruptions and ensures uninterrupted communication for both PUs and SUs. In cases where all channels are occupied, DCRA prioritizes PU access, optimizing spectrum utilization, and reducing the blocking probability for subsequent PU arrivals. Through its adaptive and responsive approach, DCRA maximizes network capacity and improves spectrum efficiency, facilitating efficient and reliable communication within the CRN ecosystem.

***Step* 3:** SU arrival

Similar to PU arrival, DCRA calculates the optimal for SUs. It is the same procedure as per the SU arrival. In this, the DCRA is used to find the optimal RC and adjust the channels for correctly occupying the SU. Then, it finds the idle channels, and the newly coming SU automatically occupies the idle channels. After that, the all channels in the CRN are filled, and at the same time SU arrives inside the channel, at that time PU gets interrupted and does not get the vacant channel in the CRN. If every channel is occupied by other SUs, newly arriving SUs will be blocked.

***Step* 4:** PU/SU departure from CRN

A PU or SU departs the CRN after completing their assignment. Then, either PUs or SUs can join the channel that has been freed up. This process helps reduce complexity and delay, managing additional traffic efficiently.

***Step* 5:** SU departure from CRN

The channel becomes available for new incoming SUs after SU finalizes its transmission. If an SU was preempted, it is allocated to this idle channel rather than being forced to terminate.

***Step* 6:** Channel failure

The total available channels in the CRN reduce when an idle channel fails, but no action is performed right away because it has no effect on existing conversations. If an occupied channel fails, spectrum handover occurs; channels are reallocated based on service priority levels for reserved and non-reserved bands.

3.3 Service Selection and Channel Restoration Using the Komodo Mlipir Optimization Algorithm

KMOA [14, 15] explains the technique for determining the appropriate service for interruption based on stated conditions. KMOA stands out for its multifaceted advantages in optimizing service selection and channel restoration within CRNs. By employing a diverse population structure and evolutionary phases, KMOA ensures adaptability to varying network conditions while maintaining scalability and efficiency. KMOA's adaptive optimization capabilities, manageable computational complexity, and global optimization prowess further underscore its versatility and applicability across a wide range of CRN scenarios, making it a potent tool for maximizing network efficiency and performance. Based on these advantages, KMOA is proposed here for Service selection and Channel restoration [16].

3.3.1 Stepwise Process of KMOA

The stepwise procedure is defined for Service selection using the KMOA Algorithm. The optimum solution is promoted using the KMOA Algorithm. The optimum solution is enhanced utilizing KMOA and the related flowchart is illustrated in Figure 2.

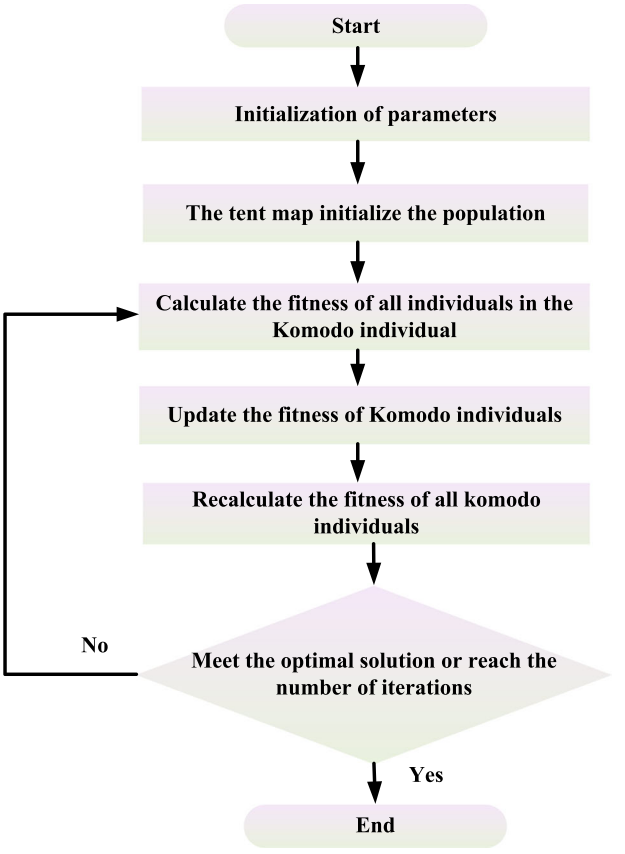


Figure 2: Flowchart for the KMO Algorithm

***Step* 1:** Initialization

The population of Komodos and the maximum count of iteration z for service selection are initialized.

***Step* 2:** Random generation

Input parameters are produced randomly after initialization. The ideal fitness value is based on their clear hyper-parameter condition.

***Step* 3:** Fitness Function

A random solution is created using initialization. It is calculated by Equation (4).

$$Fitness\,function = [service\ \ selection] \tag{4}$$

***Step* 4:** Exploration phase

The female mates with the victorious large male and give birth to two children, so expanding the viable area surrounding the global optimal solution. Parthenogenesis expands the scope of the search by providing a variety of solutions. In Equation (5), the symmetric normal distribution produces a small number at random.

$$w'_{fk} = w_{fk} + (2h-1)\beta|ij_k - gj_k| \tag{5}$$

where ij_k and gj_k signify the lower and upper bounds of kth dimension, respectively, h is a random count, β is the radius at its 0.1 value.

***Step* 5:** Exploitation phase

It is expected of tiny males to search widely for solutions and to participate in some form of exploitation as inferior people. Each small person retreats from the large men by randomly choosing a piece of their dimensions with a preset chance rate. The movement of uth Komodo to monitor b^{th} one is labeled with Milpir rate l at (0, 1) interval articulated in Equation (6)

$$w'_u = w_u + \sum_{k=1}^{z} m_{fk}, \quad where\ k \neq f \tag{6}$$

h_1, h_2 are random counts, n implies the dimension, g refers to the randomly preferred dimension, and z signifies large males. In this Milpir movement, exploration is undoubtedly more likely than exploitation.

***Step* 6:** Channel restoration

KMOA facilitates channel restoration without the need for handover, resulting in increased channel availability within the CRN.

***Step* 7:** Complexity Analysis

KMOA's computational complexity is analyzed, considering factors including the number of iterations and active Secondary Users (SUs). The algorithm's efficiency is evaluated based on its ability to handle varying traffic levels and adapt to changing network conditions.

***Step* 8:** Termination Condition

Here, the Service selection and Channel restoration depending on the KMOA Algorithm iteratively step 3 are repeated until fulfilling the $z = z + 1$ halting criterion.

3.4 Continuous Time Markov Chain (CTMC) Modeling

The suggested ESU method is modeled using CTMC discrete states. The CTMC collection of possible states is represented by E with states f. The count of fault channels is signified through g; $O_h(f)$, $O_i(f)$, and $O(f)$ denotes the count of engaged channels NR-CRN, R-CRN, and incomplete CRN. $\pi(f)$ represents steady-state probability at state (f) as expressed in Equation (7)

$$\pi * L = 0, \quad \sum_{f \in E} \pi(f) = 1 \tag{7}$$

where π denotes the steady-state probability vector, 0 denotes the row vector of every zero. The formulations for the CRN's numerous performance measures are generated in the following subsections.

3.5 Channel Availability

The channel is considered to be accessible for the service when it enters the network and discovers a channel for utilization. It is not else available. Channel availability is the possibility of a service arriving and gaining a channel. CHM_L denotes channel availability for PU as given in Equation (8)

$$CHM_L = 1 - \sum_{O(f)=J} \pi(f), \quad f \in E \tag{8}$$

where CHM_L represents PU channel availability. Channel availability for SU is expressed in Equation (9)

$$CHM_{SL} = 1 - \sum_{O(f)=J} \pi(f), \quad f \in E \tag{9}$$

where CHM_{SL} denotes the Channel availability for SU.

3.6 Service Retain Ability

Reliability quantifies the likelihood that a connection will remain operational with a given level of QoS provisioning for a predetermined period. This contrasts with the possibility of forced termination, which occurs when a service is terminated before the end of its established connection. This is stated mathematically in Equation (10)

$$SR = 1 - L^{FT} \tag{10}$$

where SR denotes the retain ability and L^{FT} signifies the service forced termination probability.

3.7 Network Unserviceable Probability (NUP)

This is the chance that a blockage, channel failure, or the arrival of a high-priority service may prevent the service from transmitting fully.

4. RESULT AND DISCUSSION

The experimental outcomes of SE-KMOA-CRN are discussed. The SE-KMOA-CRN technique is activated in MATLAB (R-2018a) using an Intel(R) Core(TM) i7-6700 CPU 3.4 GHz bearing machine. The experimental setup for validating the proposed SE-KMOA-CRN method involves simulating a CRN with 100 channels, dynamically allocated between reserved or non-reserved bands. The count of Primary Users (PUs) varies between 10 and 50, while Secondary Users (SUs) range from 20 to 100. Traffic parameters include PU arrival rates from 0.1 to 0.5 arrivals per time unit and SU arrival rates from 0.2 to 1.0 arrivals per time unit, with a fixed service rate of 1.0 for both user types. The network configuration for the SE-KMOA-CRN validation includes a CRN with a mix of 100 reserved and non-reserved channels, dynamically allocated based on real-time traffic and user demands. The CRN environment is initialized with varying numbers of PUs (10 to 50) and SUs (20 to 100), and the traffic conditions are simulated with PU arrival rates ranging from 0.1 to 0.5 arrivals per time unit and SU arrival rates from 0.2 to 1.0 arrivals per time unit. The obtained outcomes of the SE-KMOA-CRN technique are compared with the existing SUI-CRN [8], SEU-CRN [9], and SER-CRN [10] models, respectively. Table 2 shows the simulation parameter setup.

4.1 Performance Measures

It is deemed to scrutinize the effectiveness of the SE-KMOA-CRN technique.

Table 2: Simulation parameters

Parameter	Value/Range
Total Channels	100
Primary Users (PUs)	10 to 50
Secondary Users (SUs)	20 to 100
PU Arrival Rate	0.1 to 0.5 per time unit
SU Arrival Rate	0.2 to 1.0 per time unit
Service Rate	1.0 per time unit
Population Size	20 to 200

4.1.1 Throughput (TP)

This is the total number of successfully transmitted packets per unit time. Equation (11) displays the computation of performance TP.

$$TP = \frac{packet\ size}{packetarrival\ \ packetstart} \tag{11}$$

4.1.2 End-to-End Delay

The average time consumed for a packet to transfer the source to the destination is an end-end delay. This is determined by Equation (12)

$$Delay = \frac{(packet\ arrival - packet\ start)}{n\ Average\ Jitter} \tag{12}$$

4.1.3 Packet Delivery Ratio (PDR)

The amount of successfully delivered packets to total packets sent is computed by Equation (13)

$$PDR = \frac{\sum Total\ \ Number\ \ of\ packets\ \ received}{\sum Total\ \ Number\ \ of\ packets\ \ send} \times 100\% \tag{13}$$

4.1.4 Energy Efficiency (EE)

The total energy consumed per successfully transmitted packet is expressed in Equation (14)

$$EE = \frac{Total\ Energy\ Consumed}{(Total\ number\ of\ bits\ transmitted \times\ Time\ period)} \tag{14}$$

4.1.5 Spectral Efficiency (SE)

Spectral Efficiency is throughput per unit bandwidth as given in Equation (15)

$$SE = \frac{Throughput}{Bandwidth\ of\ the\ spectrum} \tag{15}$$

4.2 Performance Analysis

Figure 3–7 determines simulation outcomes of the SE-KMOA-CRN technique. Then, the SE-KMOA-CRN technique is compared with existing SUI-CRN, SEU-CRN, and SER-CRN techniques.

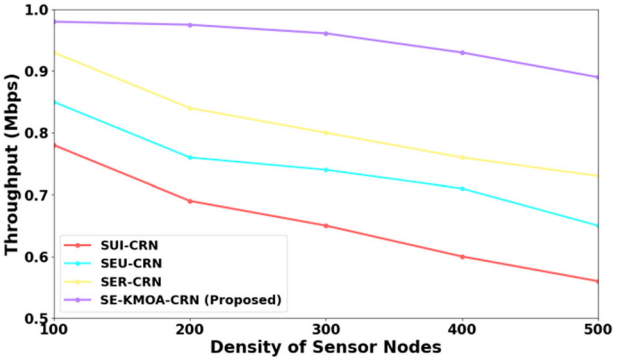


Figure 3: Throughput Analysis

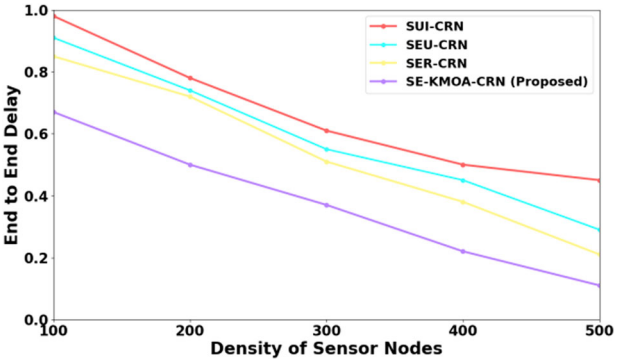


Figure 4: End-to-End Delay Analysis

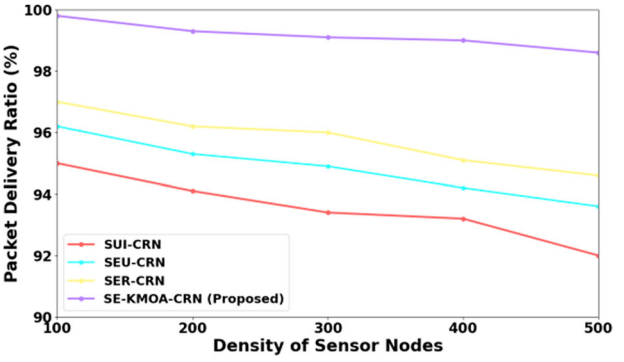


Figure 5: Packet Delivery Ratio Analysis

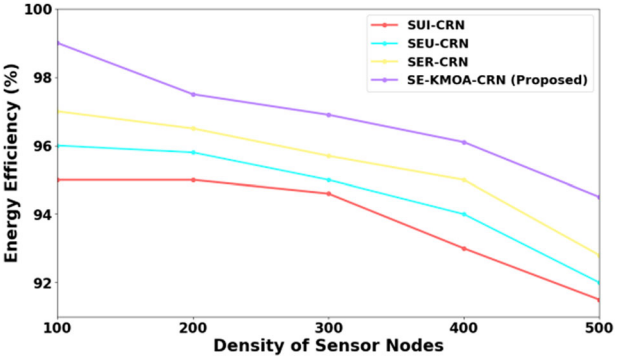


Figure 6: Energy Efficiency Analysis

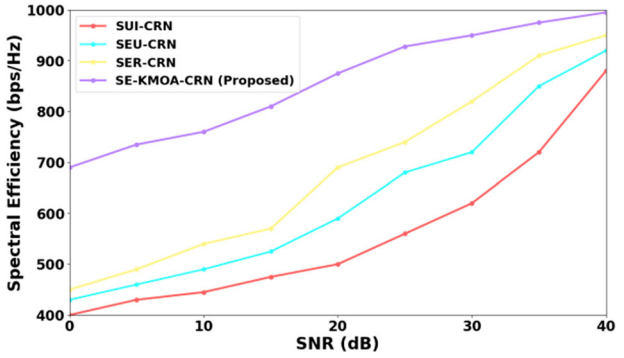


Figure 7: Spectral Efficiency Analysis

Figure 3 depicts throughput Analysis. The SE-KMOA-CRN achieves maximum throughput over the existing techniques. The SUI-CRN technique achieves the least throughput, the SER-CRN method shows slightly better outcomes with maximal throughput, and the SEU-CRN method attains moderate throughput. The proposed SE-KMOA-CRN method attains 22.42%, 29%, and 23% greater throughput compared with existing SUI-CRN, SEU-CRN, and SER-CRN techniques.

As shown in Figure 4, the SE-KMOA-CRN performs better than the existing methods by creating less latency in the network for every data packet. The SE-KMOA-CRN technique achieves the least delay with existing approaches. The SE-KMOA-CRN achieves 24.67%, 21.22%, and 25.76% less delay assessed with existing methods like SUI-CRN, SEU-CRN, and SER-CRN, respectively.

Figure 5 depicts the PDR analysis. The SE-KMOA-CRN technique achieves a maximum packet delivery ratio over the existing techniques. The SUI-CRN achieves the least packet delivery ratio, the SEU-CRN shows a slightly better outcome with maximal PDR, and the SER-CRN attains moderate PDR. The proposed SE-KMOA-CRN method attains 26.7%, 20%, and 26.4% higher PDR analyzed with existing SUI-CRN, SEU-CRN, and SER-CRN methods.

Figure 6 displays the Energy Efficiency Analysis. The SE-KMOA-CRN's energy efficiency is measured by the network lifespan. The number of packets transferred in a node is proportional to the number of requests received and data packets delivered. The average residual energy is computed regularly by summing the remaining energy in all nodes. The proposed SE-KMOA-CRN method attains 22.82%, 20.98%, and 23.73% higher energy efficiency compared with existing SUI-CRN, SEU-CRN, and SER-CRN methods.

Figure 7 depicts the Spectral Efficiency Analysis. The Spectral efficiency of the SE-KMOA-CRN is examined by the network life span. Energy consumption in a node is proportionate to the amount of packet transmissions, which is the total of requests receiving and transmitting data packets. The proposed SE-KMOA-CRN method attains 21.82%, 23.09%, and 27% greater Spectral Efficiency compared with existing SUI-CRN, SEU-CRN, and SER-CRN methods, respectively.

4.3 Discussion

The performance of the proposed SE-KMOA-CRN method was validated through extensive simulations that measured key QoS metrics, including throughput, end-end delay, energy efficacy, spectral efficiency, packet delivery ratio, and compared with existing methods like SUI-CRN, SEU-CRN, and SER-CRN, respectively. The simulations were meticulously configured to replicate realistic CRN conditions, incorporating varying traffic loads, different numbers of PUs and SUs, and dynamic channel availability. The SE-KMOA-CRN method's implementation details, including initial parameter settings, dynamic channel reservation, and service selection using KMOA, were documented to ensure consistency and reproducibility. Additionally, the simulation code and detailed instructions were made publicly available, allowing other researchers to replicate the experiments, validate results, and explore further scenarios, thereby ensuring the reliability and scalability of the proposed technique.

5. CONCLUSION

In this research, Spectrum and Energy Efficiency in CRNs using Dynamic Channel Reservation and Komodo Mlipir Optimization (SE-KMOA-CRN) is implemented successfully. Service selection and Channel restoration utilize a novel DCR and KMOA-based channel allocation. All of the measures under consideration have mathematical formulas generated using Markov chain modeling. The proposed SE-KMOA-CRN achieves 26.82%, 23.89%, and 23.09% higher throughput, 28.30%, 26.45%, and 13.48% lower End-to-End Delay, 22.08%, 25.36%, and 26.09% higher Energy Efficiency, 28.09%, 14.76%, and 26.88% greater spectral efficiency, and 18.54%, 14.02%, and 22.83% greater PDR compared with existing techniques. In the future, ESU will be combined with the hybrid underlay-interweave models of CRN. Furthermore, to check the presentation boost derived by employing QoS-based channel selection for conclusion. Finally, consider including node mobility in the model and testing the performance.

DISCLOSURE STATEMENT

No potential conflict of interest was reported by the author(s).

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